

Boyuan Zhang

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Education

Stanford University <i>M.S. in Statistics</i>	<i>Sep. 2024 – June 2026</i> (exp.)
◦ Coursework(PhD level): Causal Inference, Asymptotic and High-Dimensional Statistics, Convex Optimization, Finite Sample Statistical Theory, Theory of Probability, Applied Statistics	
New York University <i>B.A. in Computer and Data Science</i>	<i>Jan. 2021 – Dec. 2023</i>
◦ Honors: Cum Laude, University Honors Scholar, Dean's List (Academic years 2021-23) ◦ Coursework(* graduate level): Causal Inference, Mathematical Statistics*, Applied Statistics*, Probability and Statistics*, Stochastic Processes*, Statistical Learning Theory*, ML for Healthcare*, Advanced ML and Deep Learning, Data Management and Analysis, Responsible Data Science, Algorithms, Real Analysis	

Academic Interest

- Causality, Statistical Inference, Machine Learning
- Applications in healthcare, public policy, social science, and sports

Publications and Preprints

Treatment response as a latent variable Christopher Tosh, <i>Boyuan Zhang</i> , Wesley Tansey arXiv:2502.08776 	Submitted
Automated Discovery of Difference-in-Differences William Herlands, <i>Boyuan Zhang</i> , Edward McFowland III, Daniel Neill	In Preparation
A Regularized Adjusted Expected Goal Plus-Minus Model for Soccer <i>Boyuan Zhang</i> , Edvin Tran Hoac, Phung Hoang	In Preparation

Research Experience

Machine Learning for Cancer Lab <i>Memorial Sloan Kettering Cancer Center, Dept. of Epidemiology and Biostatistics</i>	<i>New York, NY</i> <i>June 2023 – May 2024</i>
◦ Research Area: Causal Testing, Empirical Bayes, Dose-response Modeling, Drug Discovery ◦ Supervisor: Dr. Wesley Tansey ◦ Developed a nonparametric compliance testing procedure for individual causal effects from observational data with potential confounding and non-compliance under an additive noise model ◦ Demonstrated effective control of the false discovery rate at the target level in simulations while maintaining reasonable power and providing more accurate estimates of the average treatment effect compared to BART, Causal Forest, and FDR Regression ◦ Discovered biologically rational markers that predispose a cell line to be sensitive or resistant to treatment with a targeted agent using data from a large cancer drug study	
Machine Learning for Good Lab <i>New York University, Courant Institute of Mathematical Science (CIMS)</i>	<i>New York, NY</i> <i>Feb. 2023 – May 2024</i>
◦ Research Area: Causal Discovery and Inference, Subset Scanning, Machine Learning ◦ Supervisor: Dr. Daniel Neill ◦ Refined and improved the subset scanning algorithms for difference-in-difference discovery ◦ Identified both heterogeneous treatment discontinuities in arbitrary-dimensional categorical data and appropriate control subsets with statistical and econometric validity ◦ Demonstrated robust performance in simulations, and discovered novel policy-relevant insights in complex	

public policy datasets

Summer Undergraduate Research Experience in Statistics

Carnegie Mellon University, Department of Statistics and Data Science

Pittsburgh, PA

June 2022 – July 2022

- Research Area: Sports Analytics, Player Evaluation, Plus-Minus Models
- Supervisors: Dr. Ronald Yurko, Dr. Konstantinos Pelechrinis
- Developed a regularized regression model for evaluating player's performance and contribution in soccer
- Incorporated priors learned from box score statistics into a regularized adjusted plus-minus model framework
- Demonstrated better predictability and interpretability than existing methods through simulations

Teaching Experience

Teaching Assistant

New York University, Center for Data Science and CIMS

New York, NY

Sep. 2022 – Dec. 2023

DS-UA 301 Advanced Machine Learning and Deep Learning

Fall 2023

DS-UA 201 Causal Inference

Summer 2023

CSCI-UA 2 Intro to Computer Programming

Fall 2022 2023, Spring 2023

Selected Projects

Analyzing the Impact of Subway Major Incidents on Passenger Flows

Summer 2023

Modeling passenger dynamics within subway network using turnstile data

Collaborative Filtering (CF) Algorithms for Recommendation Systems

Spring 2023

Comparative Analysis of Matrix Factorization, Neighborhood Models, and Neural CF

Calibrated NBA Win Probability Estimation

Spring 2023

Multi-objective RNN to estimate win probability for both accuracy and calibration

Fingertips Position Estimation of a Robot Hand

Fall 2022

CNN estimates the positions of the robot's fingertips through RGBD images

Lip Reading Word Classification

Fall 2022

CNN-LSTM-based classifier model identifies the speaker's word through video

Technologies

Languages: Python, R, SQL, Java, C/C++, MATLAB, Markdown, LaTeX

Technologies: Git, Tableau, MongoDB, Android Studio, Arduino, OCAD, Excel